

Laguerre's Inequality from Pandrosion Energy:

$$(P')^2 - P \cdot P'' = \sum Q(\alpha_k, z)^2$$

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Abstract

We prove the identity $(P'(z))^2 - P(z) \cdot P''(z) = \sum_{k=1}^d Q(\alpha_k, z)^2$, where $Q(\alpha_k, z) = \prod_{j \neq k} (z - \alpha_j)$ is the Pandrosion quotient. This holds for *all* polynomials and *all* $z \in \mathbb{C}$. For real-rooted polynomials, the right-hand side is a sum of real squares at real points, immediately giving Laguerre's inequality $(P')^2 \geq P \cdot P''$. We define the *Pandrosion energy* $E_P(z) = \sum Q(\alpha_k, z)^2$ and establish its basic properties, including the evaluation $E_P(\alpha_k) = P'(\alpha_k)^2$ and the vanishing characterisation: $E_P(z) = 0$ if and only if z is a repeated root of P .

1 The fundamental identity

Theorem 1.1 (Pandrosion–Laguerre identity). *For $P(z) = \prod_{k=1}^d (z - \alpha_k)$ with distinct or repeated roots:*

$$(P'(z))^2 - P(z) \cdot P''(z) = \sum_{k=1}^d Q(\alpha_k, z)^2, \quad (1)$$

where $Q(\alpha_k, z) = \prod_{j \neq k} (z - \alpha_j) = P(z)/(z - \alpha_k)$.

Proof. Start from the logarithmic derivative:

$$\frac{P'(z)}{P(z)} = \sum_{k=1}^d \frac{1}{z - \alpha_k}.$$

Differentiate:

$$\frac{P''(z)}{P(z)} - \left(\frac{P'(z)}{P(z)} \right)^2 = - \sum_{k=1}^d \frac{1}{(z - \alpha_k)^2}.$$

Rearranging and multiplying by $P(z)^2$:

$$\begin{aligned} (P'(z))^2 - P(z) \cdot P''(z) &= P(z)^2 \sum_{k=1}^d \frac{1}{(z - \alpha_k)^2} \\ &= \sum_{k=1}^d \left(\frac{P(z)}{z - \alpha_k} \right)^2 = \sum_{k=1}^d Q(\alpha_k, z)^2. \end{aligned} \quad \square$$

Remark 1.2. Identity (1) holds for all $z \in \mathbb{C}$, all polynomials, and all configurations of roots (including repeated ones). No hypothesis on the roots is needed.

2 Laguerre's inequality

Corollary 2.1 (Laguerre 1878). *If P has all real roots, then for every $x \in \mathbb{R}$:*

$$(P'(x))^2 \geq P(x) \cdot P''(x), \quad (2)$$

with equality if and only if x is a repeated root of P .

Proof. When $\alpha_k \in \mathbb{R}$ and $x \in \mathbb{R}$, each $Q(\alpha_k, x) = \prod_{j \neq k} (x - \alpha_j) \in \mathbb{R}$, so $Q(\alpha_k, x)^2 \geq 0$. By (1), $(P')^2 - P \cdot P'' = \sum Q^2 \geq 0$.

Equality holds iff $Q(\alpha_k, x) = 0$ for all k , i.e., x is a common zero of all $Q(\alpha_k, \cdot)$. Since $Q(\alpha_k, z)$ vanishes at α_j for $j \neq k$, the point x must be a root α_j that also kills $Q(\alpha_j, \cdot)$, which requires another root at x . Hence x is a repeated root. $\square$

3 Pandrosion energy

Definition 3.1 (Pandrosion energy). The **Pandrosion energy** of P at z is:

$$E_P(z) := \sum_{k=1}^d Q(\alpha_k, z)^2 = (P'(z))^2 - P(z) \cdot P''(z). \quad (3)$$

Proposition 3.2 (Properties of Pandrosion energy).

- (i) **Positivity:** $E_P(x) \geq 0$ for $x \in \mathbb{R}$ if P has all real roots.
- (ii) **Root evaluation:** $E_P(\alpha_k) = P'(\alpha_k)^2$ for every simple root α_k .
- (iii) **Vanishing:** $E_P(z) = 0$ if and only if z is a repeated root of P .
- (iv) **Degree:** E_P is a polynomial of degree $2(d-1)$ in z .

Proof. (i) is Corollary 2.1.

(ii) At $z = \alpha_k$: $P(\alpha_k) = 0$, so $E_P(\alpha_k) = P'(\alpha_k)^2$. Alternatively, $Q(\alpha_j, \alpha_k) = \prod_{m \neq j} (\alpha_k - \alpha_m)$: for $j = k$, $Q(\alpha_k, \alpha_k) = P'(\alpha_k)$, and for $j \neq k$, $Q(\alpha_j, \alpha_k) = 0$ (since the product contains $\alpha_k - \alpha_k = 0$). So $\sum Q(\alpha_j, \alpha_k)^2 = P'(\alpha_k)^2$.

(iii) $E_P(z) = 0$ iff $P'(z)^2 = P(z)P''(z)$, iff P and P' share the root z .

(iv) Each $Q(\alpha_k, z)$ has degree $d-1$, so Q^2 has degree $2(d-1)$, and the sum has degree $\leq 2(d-1)$. The leading coefficient is d (since each Q is monic of degree $d-1$). $\square$

Remark 3.3 (Physical interpretation). $E_P(z)$ measures the *total influence intensity* of the roots at z . Each root α_k contributes $Q(\alpha_k, z)^2 = (P(z)/(z - \alpha_k))^2$: the squared strength of the gravitational-type “pull” of α_k on the point z , weighted by the positions of all other roots. Laguerre's inequality says that the total influence is non-negative for real-rooted polynomials.

4 Connection to earlier papers

Paper	Pandrosion identity	Order
14 (Gauss–Lucas)	$P'(z) = \sum_k Q(\alpha_k, z)$	linear in Q
15 (Marden)	$Q(\alpha_k, m_k) = -(\alpha_i - \alpha_j)^2/4$	pointwise Q
18 (Walsh)	Unification via $P' = \sum Q$	linear
20 (Laguerre)	$(P')^2 - P \cdot P'' = \sum Q^2$	quadratic in Q

Paper 14 used $\sum Q$ (linear), Paper 20 uses $\sum Q^2$ (quadratic). The natural question is: what does $\sum Q^k$ give for $k \geq 3$?

Proposition 4.1 (Power-sum tower). *Define $p_m(z) := \sum_{k=1}^d Q(\alpha_k, z)^m$ (the m -th Pandrosion power sum). Then:*

- $p_1(z) = P'(z)$;
- $p_2(z) = (P'(z))^2 - P(z) \cdot P''(z)$;
- $p_m(z) = P(z)^m \sum_k (z - \alpha_k)^{-m}$, which is a polynomial of degree $m(d-1)$.

5 Sector containment

Corollary 5.1 (Sector containment of critical points). *If all roots of P lie in a sector $S = \{z : |\arg(z)| \leq \alpha\}$ with $\alpha < \pi$, then all zeros of P' lie in S .*

Proof. The sector S is convex. By Gauss–Lucas (Paper 14): if $P'(\beta) = 0$, then $\beta = \sum \lambda_k \alpha_k$ with $\lambda_k > 0$, $\sum \lambda_k = 1$, and convexity gives $\beta \in S$. $\square$

Remark 5.2. The sector theorem is a direct consequence of Gauss–Lucas (convexity of sectors). The genuinely new content of this paper is the quadratic identity (1) and the Pandrosion energy concept.

6 Conclusion

The identity $(P')^2 - P \cdot P'' = \sum Q(\alpha_k, z)^2$ elevates the Pandrosion quotient from a linear tool (Paper 14: $P' = \sum Q$) to a *quadratic* one. The “Pandrosion energy” $E_P = \sum Q^2$ provides:

- A one-line proof of Laguerre’s inequality (sum of real squares ≥ 0);
- A characterisation of repeated roots ($E_P(z) = 0$);
- A measure of “influence intensity” that interpolates between $P'(\alpha_k)^2$ at roots and the globally defined polynomial $E_P(z)$.

References

- [1] E. Laguerre, *Sur la théorie des équations numériques*, J. Math. Pures Appl. (3) **9** (1883), 99–146.
- [2] I. Besevic, *Derivative-free proofs of Gauss–Lucas via Pandrosion influence weights*, Preprint (2026).